

## Practical – 01

Code:

```
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>
#include <string.h>
int main()
{
    char command[100];
    char *args[20];
    printf("Enter Linux command: ");
    fgets(command, sizeof(command), stdin);
    command[strcspn(command, "\n")] = '\0';
    int i = 0;
    args[i] = strtok(command, " ");
    while(args[i] != NULL)
    {
        i++;
        args[i] = strtok(NULL, " ");
    }
    pid_t pid = fork();
    if(pid == 0)
    {
        printf("\nChild Process\n");
        printf("Child PID : %d\n", getpid());
        printf("Parent PID: %d\n\n", getppid());
        execvp(args[0], args);
        perror("Execution Failed");
        exit(1);
    }
```

```
GNU nano 7.2
        i++;
        args[i] = strtok(NULL, " ");
    }
    pid_t pid = fork();
    if(pid == 0)
    {
        printf("\nChild Process\n");
        printf("Child PID : %d\n", getpid());
        printf("Parent PID: %d\n\n", getppid());
        execvp(args[0], args);
        perror("Execution Failed");
        exit(1);
    }
    else if(pid > 0)
    {
        printf("\nParent Process\n");
        printf("Parent PID: %d\n", getpid());
        printf("Child PID : %d\n", pid);
        wait(NULL);
        printf("\nChild execution completed.\n");
    }
    else
    {
        printf("Fork failed.\n");
    }
    return 0;
}
```

Output:

If input – (ls -l):

```
bharath_sai_reddy@LAPTOP-1711DF88:~/90128_OSSP/2520090128_Practical/Practical-01$ ./command_execution
Enter Linux command: ls -l

Parent Process
Parent PID: 698
Child PID : 699

Child Process
Child PID : 699
Parent PID: 698

command_execution
command_execution.c

Child execution completed.
bharath_sai_reddy@LAPTOP-1711DF88:~/90128_OSSP/2520090128_Practical/Practical-01$
```

If input – (uname):

```
bharath_sai_reddy@LAPTOP-1711DF88:~/90128_OSSP/2520090128_Practical/Practical-01$ ./command_execution
Enter Linux command: uname

Parent Process
Parent PID: 727
Child PID : 738

Child Process
Child PID : 738
Parent PID: 727

Linux

Child execution completed
```

If input – (lscpu):

```
bharath_sai_reddy@LAPTOP-1711DF88:~/90128_OSSP/2520090128_Practical/Practical-01$ ./command_execution
Enter Linux command: lscpu

Parent Process
Parent PID: 762
Child Process
Child PID : 763
Child PID : 763
Parent PID: 762

Architecture:                x86_64
CPU op-mode(s):              32-bit, 64-bit
Address sizes:               39 bits physical, 48 bits virtual
Byte Order:                  Little Endian
CPU(s):                      12
On-line CPU(s) list:        0-11
Vendor ID:                   GenuineIntel
Model name:                  13th Gen Intel(R) Core(TM) i5-13420H
CPU family:                  6
Model:                       186
Thread(s) per core:          2
Core(s) per socket:          6
Socket(s):                   1
Stepping:                    2
BogoMIPS:                    5222.40
Flags:                        fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clflush mmx fxsr sse sse2 ss h
t syscall nx pdpe1gb rdtscp lm constant_tsc rep_good nopl xtopology tsc_reliable nonstop_tsc cpuid tsc
_known_freq pni pclmulqdq vmx ssse3 fma cx16 pcid sse4_1 sse4_2 x2apic movbe popcnt tsc_deadline_timer
aes xsave avx f16c rdrand hypervisor lahf_lm abm 3dnowprefetch ssbd ibrs ibpb stibp ibrs_enhanced tpr
_shadow ept vpid ept_ad fsgsbase tsc_adjust bmi1 avx2 smep bmi2 erms invpcid rdseed adx smap clflushop
t clwb sha_ni xsaveopt xsavec xgetbv1 xsaves avx_vnni vnni umip waitpkg gfni vaes vpclmulqdq rdpid mov
diri movdir64b fsrm md_clear serialize ibt flush_l1d arch_capabilities

Virtualization features:
Virtualization:              VT-x
Hypervisor vendor:           Microsoft
```

```

Hypervisor vendor:    Microsoft
Virtualization type:  full
Caches (sum of all):
  L1d:                288 KiB (6 instances)
  L1i:                192 KiB (6 instances)
  L2:                 7.5 MiB (6 instances)
  L3:                 12 MiB (1 instance)
NUMA:
  NUMA node(s):       1
  NUMA node0 CPU(s):  0-11
Vulnerabilities:
  Gather data sampling: Not affected
  Ghostwrite:          Not affected
  Indirect target selection: Not affected
  Itlb multihit:       Not affected
  L1tf:                Not affected
  Mds:                 Not affected
  Meltdown:            Not affected
  Mmio stale data:     Not affected
  Old microcode:       Not affected
  Reg file data sampling: Mitigation; Clear Register File
  Retbleed:            Mitigation; Enhanced IBRS
  Spec rstack overflow: Not affected
  Spec store bypass:   Mitigation; Speculative Store Bypass disabled via prctl
  Spectre v1:          Mitigation; usercopy/swapgs barriers and __user pointer sanitization
  Spectre v2:          Mitigation; Enhanced / Automatic IBRS; IBPB conditional; PBRSE-eIBRS SW sequence; BHI BHI_DIS_S
  Srbds:               Not affected
  Tsa:                 Not affected
  Tsx async abort:     Not affected
  Vmscape:             Not affected

Child execution completed.

```

If input – (lsblk):

```

bharath_sai_reddy@LAPTOP-1711DF88:~/90128_OSSP/2520090128_Practical/Practical-01$ ./command_execution
Enter Linux command: lsblk

Parent Process
Parent PID: 779
Child Process
Child PID : 782
Child PID : 782
Parent PID: 779

NAME
MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
sda  8:0    0 356.9M 1 disk
sdb  8:16   0 159.4M 1 disk
sdc  8:32   0    2G  0 disk [SWAP]
sdd  8:48   0    1T  0 disk /mnt/wslg/distro
/

Child execution completed.

```

If input – (ps):

```

bharath_sai_reddy@LAPTOP-1711DF88:~/90128_OSSP/2520090128_Practical/Practical-01$ ./command_execution
Enter Linux command: ps

Parent Process
Parent PID: 787
Child PID : 788

Child Process
Child PID : 788
Parent PID: 787

  PID TTY          TIME CMD
  315 pts/0        00:00:00 bash
  787 pts/0        00:00:00 command_executi
  788 pts/0        00:00:00 ps

Child execution completed.

```

If input – (top):

```
bharath_sai_reddy@LAPTOP-1711DF88:~/90128_OSSP/2520090128_Practical/Practical-01$ ./command_execution
Enter Linux command: top
```

```
Parent Process
Parent PID: 817
Child PID : 818
```

```
Child Process
Child PID : 818
```

```
Parent PID: 817
```

```
top - 03:19:22 up 31 min, 1 user, load average: 0.06, 0.03, 0.00
```

```
Tasks: 23 total, 1 running, 22 sleeping, 0 stopped, 0 zombie
```

```
%Cpu(s): 0.0 us, 0.0 sy, 0.0 ni, 100.0 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
```

```
Mem: 7756.6 total, 7145.2 free, 514.3 used, 247.7 buff/cache
```

```
Swap: 2048.0 total, 2048.0 free, 0.0 used, 7242.3 avail Mem
```

| PID | USER     | PR | NI | VIRT   | RES   | SHR   | S | %CPU | %MEM | TIME+   | COMMAND         |
|-----|----------|----|----|--------|-------|-------|---|------|------|---------|-----------------|
| 1   | root     | 20 | 0  | 21856  | 13152 | 9796  | S | 0.0  | 0.2  | 0:01.09 | systemd         |
| 2   | root     | 20 | 0  | 3180   | 2212  | 2072  | S | 0.0  | 0.0  | 0:00.02 | init-systemd(Ub |
| 6   | root     | 20 | 0  | 3180   | 2048  | 1940  | S | 0.0  | 0.0  | 0:00.00 | init            |
| 42  | root     | 19 | -1 | 66760  | 19504 | 18372 | S | 0.0  | 0.2  | 0:00.52 | systemd-journal |
| 93  | root     | 20 | 0  | 25412  | 6704  | 5144  | S | 0.0  | 0.1  | 0:00.78 | systemd-udevd   |
| 107 | systemd+ | 20 | 0  | 21344  | 13184 | 11060 | S | 0.0  | 0.2  | 0:00.14 | systemd-resolve |
| 122 | systemd+ | 20 | 0  | 91036  | 7980  | 7008  | S | 0.0  | 0.1  | 0:00.14 | systemd-timesyn |
| 168 | root     | 20 | 0  | 4244   | 2684  | 2432  | S | 0.0  | 0.0  | 0:00.01 | cron            |
| 169 | message+ | 20 | 0  | 9544   | 5348  | 4644  | S | 0.0  | 0.1  | 0:00.14 | dbus-daemon     |
| 185 | root     | 20 | 0  | 17980  | 8884  | 7836  | S | 0.0  | 0.1  | 0:00.13 | systemd-logind  |
| 201 | syslog   | 20 | 0  | 222516 | 5904  | 4500  | S | 0.0  | 0.1  | 0:00.09 | rsyslogd        |
| 209 | root     | 20 | 0  | 3124   | 1976  | 1836  | S | 0.0  | 0.0  | 0:00.00 | agetty          |
| 215 | root     | 20 | 0  | 112820 | 25088 | 15136 | S | 0.0  | 0.3  | 0:00.21 | unattended-upgr |
| 313 | root     | 20 | 0  | 3184   | 1116  | 980   | S | 0.0  | 0.0  | 0:00.00 | SessionLeader   |
| 314 | root     | 20 | 0  | 3200   | 1252  | 1108  | S | 0.0  | 0.0  | 0:00.09 | Relay(315)      |
| 315 | bharath+ | 20 | 0  | 6080   | 5388  | 3664  | S | 0.0  | 0.1  | 0:00.12 | bash            |
| 316 | root     | 20 | 0  | 6704   | 4664  | 3880  | S | 0.0  | 0.1  | 0:00.02 | login           |
| 364 | bharath+ | 20 | 0  | 20120  | 11388 | 9452  | S | 0.0  | 0.1  | 0:00.23 | systemd         |
| 365 | bharath+ | 20 | 0  | 21160  | 3592  | 1868  | S | 0.0  | 0.0  | 0:00.00 | (sd-pam)        |